LECTURE 35: MATH1061/7861
TOPICS: THE PIGEONHOLE PRINCIPLE, INTRODUCTION TO GRAPH THEORY
DATE: WEDNESDAY MAY 20, 2026

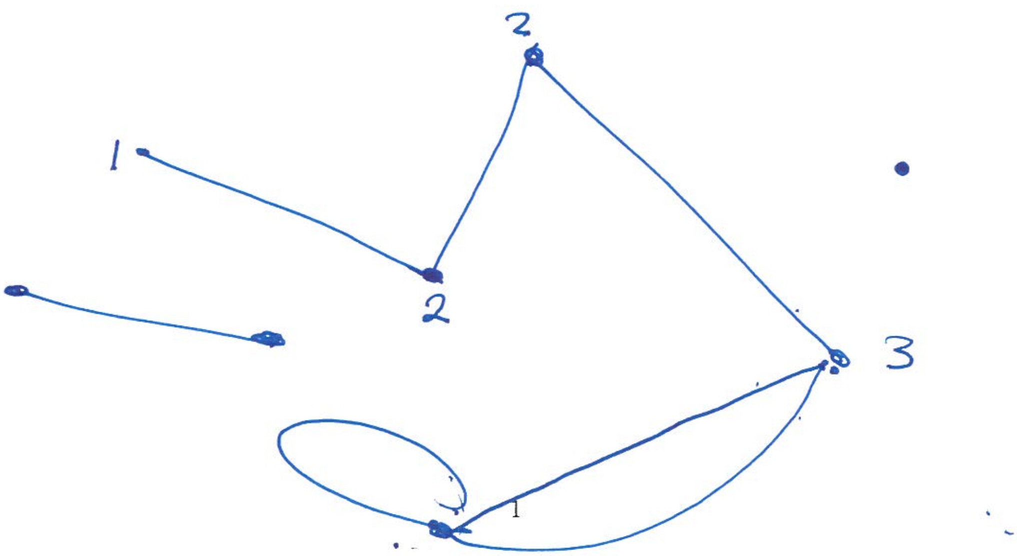
Learning objectives:

(1) Learn the Pigeonhole Principle and use it to solve problems.

(2) Understand basic definitions and notation in graph theory.

□ □ □ □ .... □ n pidgeonholes.

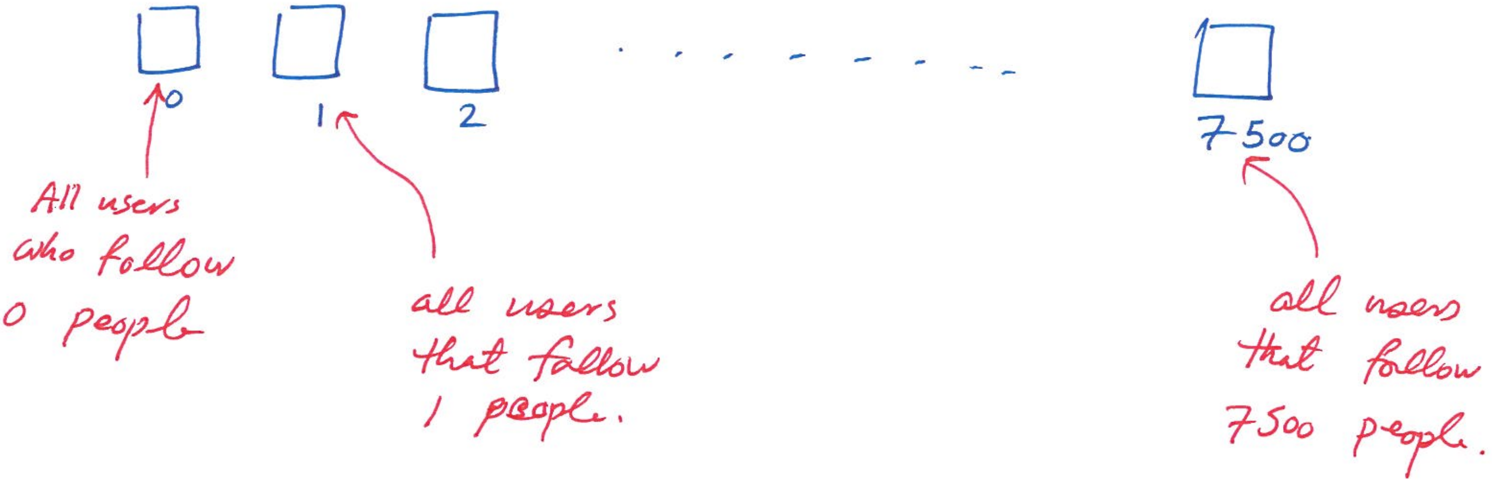
number of pigeons $> n$, then there is at least one pigeonhole which contain ≥ 2 pigeons.



Q1: There are 2 billion people using Instagram. The maximum number of people a user can follow is 7500.

True of false: There exists a number $k \in \{0, 1, 2, \ldots, 7500\}$ such that the number of Instagram users who follow exactly k people is $> 200,000$.

Pigeonholes:



The statement is true.

Suppose, for the sake of contradiction, that the statement was false. So then each pigeonhole has $\leq 200,000$ pigeons.

So we have 7501 pigeonhole, so total number of pigeons must be $\leq \underline{7501 \times 200,000}$

1,500,200,000

< 2 billion.

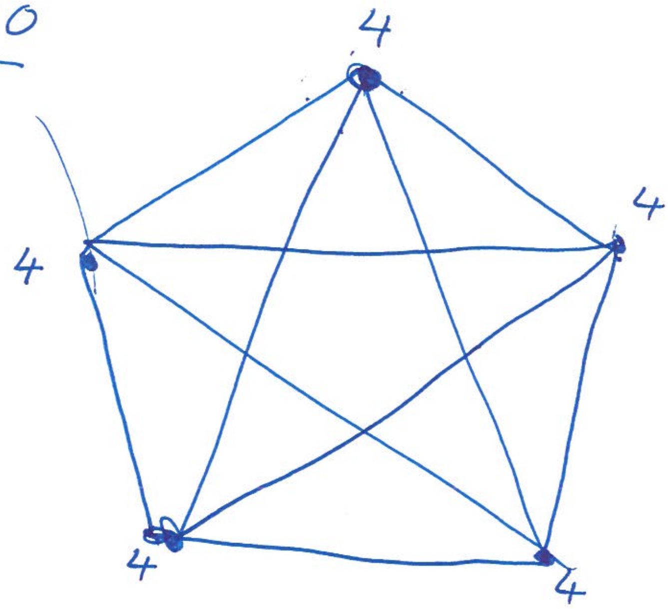
Contradictions.

Q2: You have 5 computers in a network. Each computer can connect ~~directly~~ to any other computer.

(a) What is the maximum number of connections possible?

Draw the graph and compute the degrees of the vertices and the number of edges.

edges = 10



$\boxed{\text{Complete Graphs.}}$

$$4 + 3 + 2 + 1 = 10$$

$$(n-1) + (n-2) + \cdots + 1 = \frac{n(n-1)}{2} = \binom{n}{2}.$$

(b) Suppose we want every computer to be able to communicate with every other computer. What is the minimum umber of connections required?

Draw the graph and compute the degrees of the vertices and the number of edges.